

AI Regulations

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Topics to be covered

- Importance of AI Regulations
- Regulatory landscape
- Essential Elements of the EU AI Act
 - Subject matter and scope
 - Terminology
 - Prohibited AI systems
 - High Risk AI systems
- Requirements for high risk AI system
- Penalties for non-compliance

Importance of AI regulations

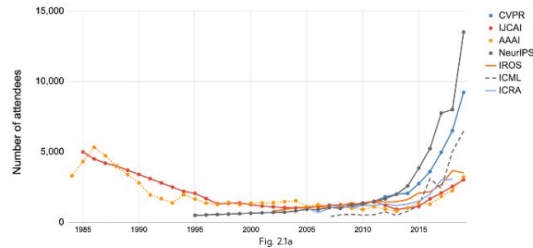
Automation Bias

- If computers have done it, it must be right
- If AI has done it, it must be right
 - However, AI is still emerging!
- From Wikipedia “the propensity for humans to favor suggestions from automated decision-making systems and to ignore contradictory information made without automation, even if it is correct.”

Exponential Growth of AI vs Legal Interia

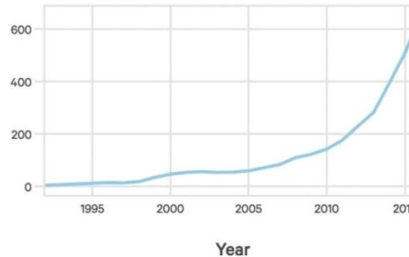
- The speed at which AI is evolving gives too little time for the society to respond
 - Implying legal backwardness

Attendance at large conferences (1984-2019)
Source: Conference provided data.



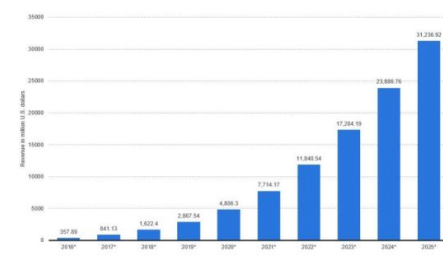
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

Source: Crunchbase, VentureSource, Sand
Hill Econometrics



**Enterprise Application AI
Revenue**

Source: Statista

AI Impact

- This exponential growth coupled with insufficient checks may be catastrophic
 - May impact
 - Health
 - Safety
 - Fundamental Rights
 - Democracy
 - Rule of Law
 - Environment

Negative Impact on Health

- AI chatbots like ChatGPT giving wrong medical advice

AI Shocker: Man Dies By Suicide After Six-Day Chatting With ChatGPT-like Chatbot

Reports claimed that the man referred to as Pierre took his own life after talking to AI bot ELIZA for six weeks on climate change. However, the conversation about the climate issues of the world turned 'confusing and harmful', prompting him to end his own life.

Times Now Bureau, Surabhi Shaurya | Updated Mar 31, 2023, 06:13 PM IST



WORLD

AI bots like ChatGPT can now influence people in taking decisions related to life and death

Experts are calling for future bots to be banned from giving advice on ethical issues, after researchers found that AI bots can seriously influence people and the decisions they take even in life and death situations, in a haphazard manner.

Mehul Reuben Das | April 07, 2023 10:36:41 IST



Source:

<https://www.timesnownews.com/technology-science/ai-shocker-man-dies-by-suicide-after-six-day-chatting-with-chatgpt-like-chatbot-article-99146570>

<https://www.firstpost.com/world/ai-bots-like-chatgpt-can-influence-people-in-decisions-related-to-life-and-death-12421072.html>

Violation of fundamental rights

- E.g. right to equality

Artificial Intelligence and Gender Equality: Is AI a Friend or Foe of Inclusion?



Alessandro Mascellino
Technology Journalist

Last updated: 26 October, 2023



Disclosure



Why Trust Us



KEY TAKEAWAYS

Uncover AI's gender gap bridging potential by erasing biases in recruitment and pay, amid bias encoding challenges. Explore inclusive skill and leadership initiatives, addressing ethical concerns for a diverse, equitable tech-driven future.

Potential to Impact Democracy!

- Possible to generate huge amount of misinformation
 - at virtually no cost!
- An era in which nobody trusts anything!

AI-generated disinformation poses threat of misleading voters in 2024 election

Politics May 14, 2023 7:52 PM EST

WASHINGTON (AP) — Computer engineers and tech-inclined political

A.I. Muddies Israel-Hamas War in Unexpected Way

Fakes related to the conflict have been limited and largely unconvincing, but their presence has people doubting real evidence.

Source:

<https://www.pbs.org/newshour/politics/ai-generated-disinformation-poses-threat-of-misleading-voters-in-2024-election>
<https://www.nytimes.com/2023/10/28/business/media/ai-muddies-israel-hamas-war-in-unexpected-way.html>

Threatens Rule of Law!

- Denial of genuine evidences in a court of law

WATERLOO REGION

AI threatens courts with fake evidence, UW prof says

Ethics prof supports pause in development and deployment of AI until safeguards are in place

 By **Terry Pender** Record Reporter
Tuesday, March 28, 2023 | ⌚ 2 min to read

Source:

https://www.therecord.com/news/waterloo-region/ai-threatens-courts-with-fake-evidence-uw-prof-says/article_a94e841a-b603-59bc-85fb-24a5ed044e2a.html

Serious impact on the Environment

- GPT-4 training
 - \$100 million
 - 100 days
 - 25,000 NVIDIA A100 GPUs
 - Server with these GPUs use about 6.5 kW each
 - An estimated 50 GWh!
- The estimated energy consumption of a GPT-4 query is 0.001-0.01 kWh (3.6-36 kJ)
- Increased e-waste!

Source:

<https://www.ri.se/en/news/blog/generative-ai-does-not-run-on-thin-air>

<https://lifestyle.livemint.com/news/big-story/ai-carbon-footprint-openai-chatgpt-water-google-microsoft-111697802189371.html>

<https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/#:~:text=This%20is%20a%20time%2Dconsuming,San%20Francisco%20for%20three%20days>

AI in recruitment and selection

https://www.theregister.com/2020/12/08/texas_compsci_phd_ai/

The RegisterOFF-PREM ▼ ON-PREM ▼ SOFTWARE ▼ SECURITY OFF-BEAT ▼ VENDOR VOICE ▼Q

{* ARTIFICIAL INTELLIGENCE *}

Uni revealed it killed off its PhD-applicant screening AI – just as its inventors gave a lecture about the tech

Fears of bias put compsci dept into damage-limitation mode after years of using it to analyze applications

Katyanna Quach Tue 8 Dec 2020 // 12:04 UTC SHARE

A university announced it had ditched its machine-learning tool, used to filter thousands of PhD applications, right as the software's creators were giving a talk about the code and drawing public criticism.

// MOST READ



Apple fires warning shot at Facebook and Google on privacy, pledges fight

Source: https://www.theregister.com/2020/12/08/texas_compsci_phd_ai/

AI in policing

Predictive policing poses discrimination risk, thinktank warns

Machine-learning algorithms could replicate or amplify bias on race, sexuality and age



▲ One officer said human biases including more stop and searches of black men were likely to be introduced into algorithm data sets. Photograph: Carl Court/Getty Images

Two Drug Possession Arrests



DYLAN FUGETT

LOW RISK

3



BERNARD PARKER

HIGH RISK

10

Fugett was rated low risk after being arrested with cocaine and marijuana. He was arrested three times on drug charges after that.

Already voices against the use of AI in policing!

Why Hundreds of Mathematicians Are Boycotting Predictive Policing

Some academics are calling the controversial practice a "scientific veneer for racism."



// BY COURTNEY LINDER JUL 20, 2020

Los Angeles Times

   LAPD pioneered predicting crime with data. Many police don't think it works

Source:
<https://www.popularmechanics.com/science/math/a32957375/mathematicians-boycott-predictive-policing/>
<https://www.latimes.com/local/lanow/la-me-lapd-precision-policing-data-20190703-story.html>

Other significant areas

- Protection of Intellectual Property Rights
- Deepfakes for pornography
- Chatbots glorifying suicide
-

Regulatory Landscape

Major Approaches to AI Regulation

- The Comprehensive Model
 - Example, European Union's AI Act
- The Sector-specific Model
 - Primary actor is the United States
 - No single, overarching federal AI law
 - Different agencies for different sectors
 - FTC, FDA, EEOC, etc.
 - NIST AI Risk Management Framework
 - Some states regulating AI in the US
 - California, Colorado, etc.
- China model
 - State governance, state control

Our Focus: EU AI Act

The Purpose

- Improve the functioning of the internal market
- Promote the uptake of human centric and trustworthy AI

While ensuring

- A high level of protection of
 - Health
 - Safety
 - Fundamental rights
 - Democracy
 - Rule of law
 - Environment

Against the harmful effects of AI

- Supporting innovation

Main Actors

Provider

- natural or legal person
- public authority
- agency or other body
 - that develops an AI system or a general-purpose AI model or
 - that has an AI system or a general-purpose AI model developed
- and
 - places it on the market or
 - puts the AI system into service under its own name or trademark,
 - whether for payment or free of charge;

Deployer

- natural or legal person
- public authority,
- agency or other body
 - using an AI system
 - under its authority
- **except**
 - where the AI system is used in the course of a personal non-professional activity;

Authorised Representative

- natural or legal person
 - located or established in the Union
 - who has received and accepted
 - a written mandate
 - from a provider of an AI system or a general-purpose AI model
 - to perform and carry out on its behalf the obligations and procedures established by this Regulation;

Main Actors in the EU AI Act

- ‘importer’ means a natural or legal person located or established in the Union that places on the market an AI system that bears the name or trademark of a natural or legal person established in a third country;
- ‘distributor’ means a natural or legal person in the supply chain, other than the provider or the importer, that makes an AI system available on the Union market
- ‘operator’ means a provider, product manufacturer, deployer, authorised representative, importer or distributor;

Subject Matter

- harmonised rules for
 - the placing on the market,
 - the putting into service, and
 - the use of AI systems in the Union;

- supply of an AI system
- for first use directly to the deployer
- or for own use in the Union for its intended purpose

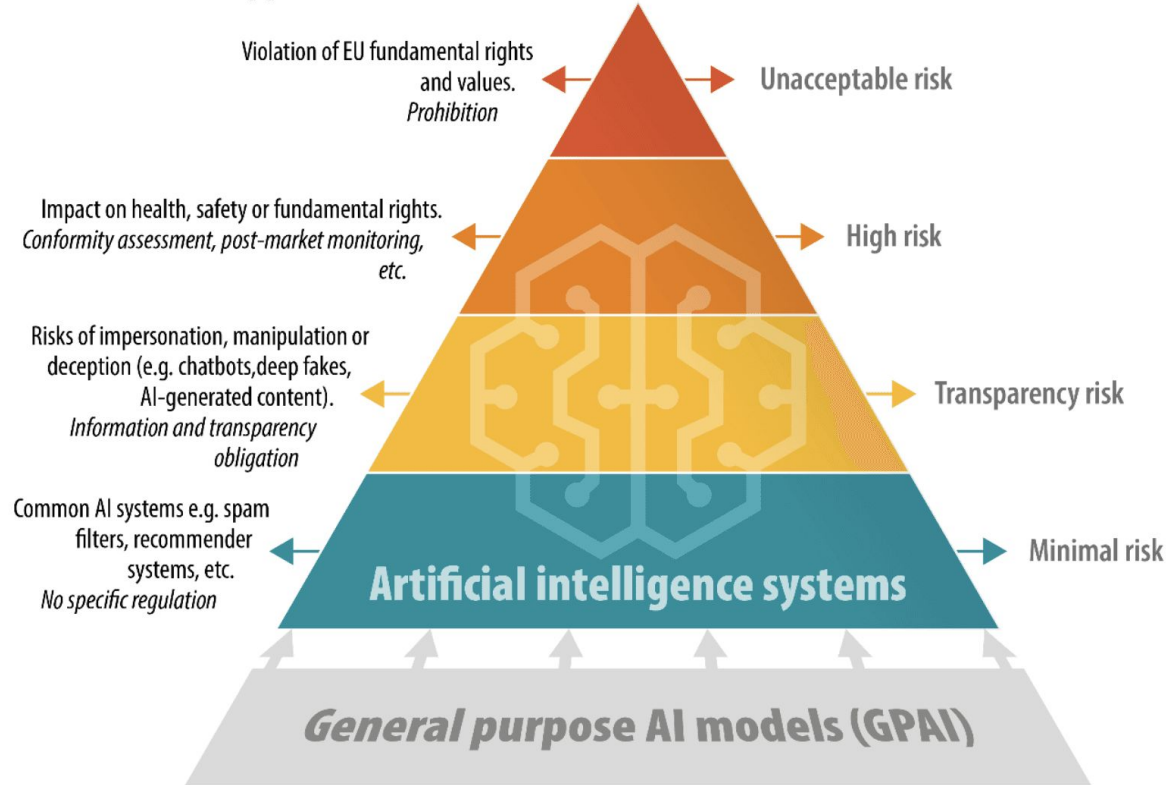
- first making available
- AI system or a general-purpose AI model
- on the Union market

- the supply of
- AI system or a general-purpose AI model
- for distribution or use
- on the Union market
- for commercial activity,
- whether in return for payment or free of charge;

Subject Matter: The Risk Pyramid

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Advocate, Bar Council of Rajasthan.

EU AI act risk-based approach



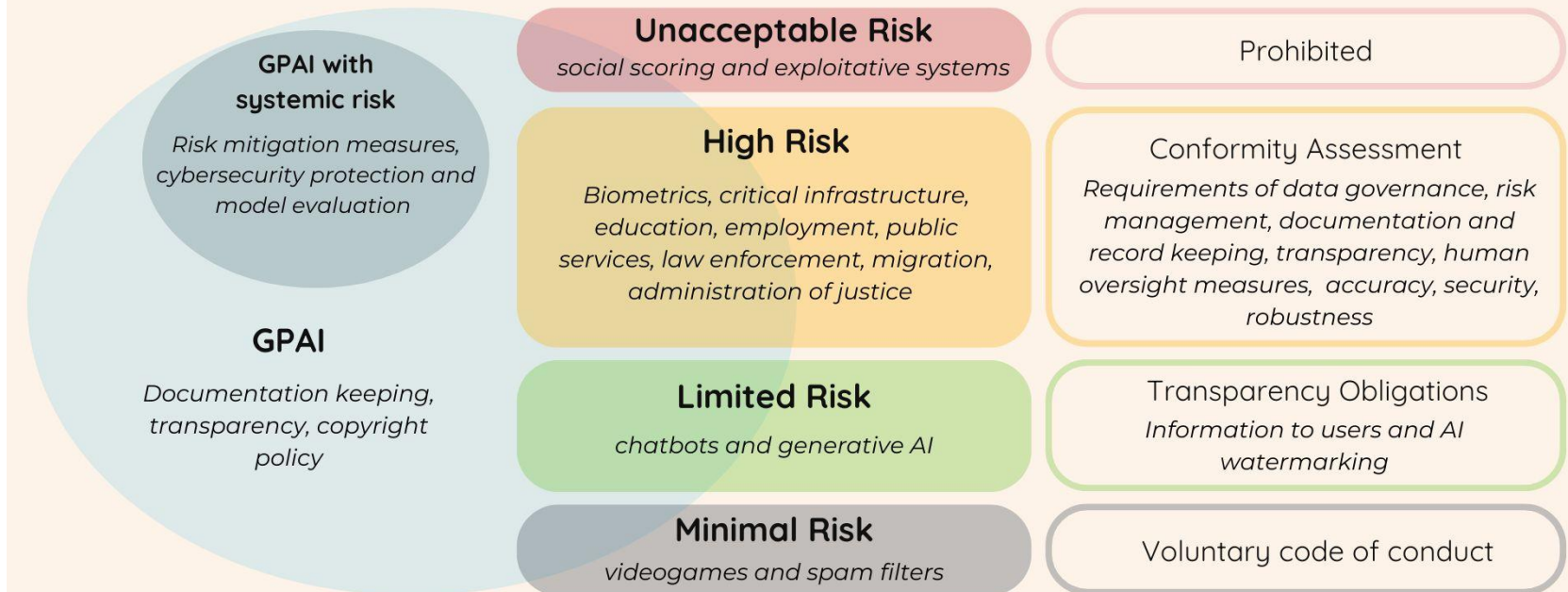
GPAI models - *Transparency requirements*

GPAI with systemic risks - *Transparency requirements, risk assessment and mitigation*

Subject Matter: The Risk Pyramid

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THE EU AI ACT RISK LEVELS



© Francesca Palmiotto

Originally posted by Francesca Palmiotto at: https://www.linkedin.com/posts/francesca-palmiotto-672a76b6_euaiact-ai-activity-7265011527177404416-L1Qe/

Subject Matter: Governance and supporting innovation

- Rules on
 - market monitoring,
 - market surveillance,
 - governance and enforcement;
- Measures to support innovation,
 - with a particular focus on SMEs,
 - including start-ups

General purpose AI model

- trained with a large amount of data
 - using self-supervision at scale
 - that displays significant generality
 - capable of competently performing a wide range of distinct tasks
 - regardless of the way the model is placed on the market
 - can be integrated into a variety of downstream systems or applications
- except
- AI models that are used for research, development or prototyping activities before they are placed on the market;

Definitions related to GPAI

- ‘general-purpose AI model’ means an AI model, including where such an AI model is trained with a large amount of data using self-supervision at scale, that displays significant generality and is capable of competently performing a wide range of distinct tasks regardless of the way the model is placed on the market and that can be integrated into a variety of downstream systems or applications, except AI models that are used for research, development or prototyping activities before they are placed on the market;
- ‘general-purpose AI system’ means an AI system which is based on a general-purpose AI model and which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems;

Scope

1. This Regulation applies to:
 - (a) providers placing on the market or putting into service AI systems *or placing on the market general-purpose AI models* in the Union, irrespective of whether those providers are established *or located* within the Union or in a third country;
 - (b) *deployers* of AI systems *that have their place of establishment or are* located within the Union;
 - (c) providers and *deployers* of AI systems that *have their place of establishment or are* located in a third country, where the output produced by the AI system is used in the Union;
 - (d) *importers and distributors of AI systems;*
 - (e) *product manufacturers placing on the market or putting into service an AI system together with their product and under their own name or trademark;*
 - (f) *authorised representatives of providers, which are not established in the Union;*
 - (g) *affected persons that are located in the Union.*

This Regulation applies to

- providers
 - placing on the market or
 - putting into service AI systems or
 - placing on the market general-purpose AI models in the Union
 - irrespective of whether established or located within the Union or in a third country

This Regulation applies to

- **deployers**
 - of AI systems that have their place of establishment or are located within the Union;



This Regulation applies to

- providers and deployers of AI systems
 - that have their place of establishment or are located in a **third country**
 - where the output produced by the AI system is **used** in the Union;

Does not apply

- Military, defense or national security purposes
- Sole purpose of scientific research and development
- To any research, testing or development activity
 - Prior
 - In accordance with the EU law
 - Testing in real world not covered
- To natural persons using AI systems in the course of a purely personal non-professional activity
- AI systems released under free and open-source licences
 - Exceptions for that fall in the first three categories of the pyramid

Does not apply

- Outside the scope of EU law
- Public authorities in a third country, nor to international organizations
- So as to decrease consumer rights and safety
- So as to preclude member states from making laws favourable to workers

Article 4: AI literacy

Providers and deployers of AI systems shall take measures to ensure, to their best extent, a sufficient level of AI literacy of their staff and other persons dealing with the operation and use of AI systems on their behalf, taking into account their technical knowledge, experience, education and training and the context the AI systems are to be used in, and considering the persons or groups of persons on whom the AI systems are to be used.

Prohibited AI

Prohibited AI

- Malicious use of subliminal techniques
- Maliciously exploits any of the vulnerabilities of a natural person or a specific group of persons
 - due to their age, disability or a specific social or economic situation
- Evaluation or classification of natural persons or groups of persons
 - Social scoring
- Making risk assessments of natural persons
 - to assess or predict the risk of a natural person committing a criminal offence
 - Based **solely** on the **profiling** of a natural person or on assessing their personality traits and characteristics
- AI systems that create or expand facial recognition databases through the **untargeted** scraping of **facial images** from the Internet or CCTV footage;

Prohibited AI

- AI systems to **infer emotions** of a natural person in the areas of workplace and education institutions
- Use of **biometric categorisation systems** that categorise individually natural persons based on their biometric data to deduce or infer their
 - race
 - political opinions
 - trade union membership
 - religious or philosophical beliefs
 - sex life or sexual orientation
- use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement

High Risk AI

High Risk AI

- **Biometrics**
 - Remote biometric identification systems
 - Biometric categorisation
 - Emotion recognition
- **Critical Infrastructure**
 - critical digital infrastructure, road traffic, or in the supply of water, gas, heating or electricity
- **Educational and Vocational Training**
 - Determine access or admission to educational institutions
 - Evaluate learning outcomes
 - Monitoring and detecting prohibited behaviour during exams/tests

High Risk AI

- Employment, workers management and access to self-employment
 - Recruitment or selection of natural persons
 - In promotions or termination or appraisals, etc
- Access to and enjoyment of essential private services and essential public services and benefits
 - To evaluate eligibility of natural persons for such services
 - To evaluate creditworthiness of natural persons
 - Risk assessment and pricing in case of life and health insurance
 - Evaluate and classify emergency calls by natural persons

High Risk AI

- Law enforcement
 - to assess the risk of a natural person becoming the victim of criminal offences
 - polygraphs or similar tools
 - to evaluate the reliability of evidence in the course of the investigation or prosecution of criminal offences
 - for assessing the risk of a natural person offending or re-offending
 - to assess personality traits and characteristics or past criminal behaviour of natural persons or groups
 - for the profiling of natural persons in the course of the detection, investigation or prosecution of criminal offences

High Risk AI

- Migration, asylum and border control management
 - polygraphs or similar tools
 - to assess a risk, including a security risk, a risk of irregular migration, or a health risk, posed by a natural person who intends to enter or who has entered into the territory of a Member State
 - for the examination of applications for asylum, visa or residence permits and for associated complaints with regard to the eligibility of the natural persons applying for a status, including related assessments of the reliability of evidence
 - detecting, recognising or identifying natural persons
 - exception of the verification of travel documents

High Risk AI

- Administration of justice and democratic processes
 - researching and interpreting facts and the law and in applying the law to a concrete set of facts
 - Or in alternative dispute resolution
 - AI systems intended to be used for influencing the outcome of an election or referendum or the voting behaviour of natural persons in the exercise of their vote in elections or referenda

Requirements for High Risk AI Systems

Risk Management System

- A risk management system shall be established, implemented, documented and maintained
- A continuous iterative process planned and run throughout the entire lifecycle
 - the identification and analysis of the known and the reasonably foreseeable risks
 - the estimation and evaluation of the risks that may emerge
 - the evaluation of other risks possibly arising, based on the analysis of data gathered from the post-market monitoring
 - the adoption of appropriate and targeted risk management measures

Data and data governance

- Training, validation and testing data sets
 - All three a **legal** requirement now
 - Subject to data governance practices
- For example,
 - an assessment of the availability, quantity and suitability of the data sets that are needed
 - examination in view of possible biases that are likely to affect the **health**
- Training, validation and testing data sets shall be
 - relevant
 - sufficiently representative
 - to the best extent possible, free of errors
 - complete in view of the intended purpose
- Data sets shall take into account the specific geographical, contextual, behavioural or functional setting

Technical documentation

- The technical documentation of a high-risk AI system shall be drawn up
 - before that system is placed on the market or
 - put into service and
 - shall be kept up-to date

Record keeping

- High-risk AI systems shall technically allow for the automatic recording of events (logs) over the lifetime of the system

Transparency and provision of information to deployers

- High-risk AI systems shall be designed and developed in such a way as to ensure that their operation is sufficiently transparent to enable deployers to interpret a system's output and use it appropriately.

Human Oversight

- High-risk AI systems shall be designed and developed in such a way, including with appropriate human-machine interface tools, that they can be effectively overseen by natural persons during the period in which they are in use.

Accuracy, robustness and cybersecurity

- High-risk AI systems shall be designed and developed in such a way that they achieve an appropriate level of accuracy, robustness, and cybersecurity, and that they perform consistently in those respects throughout their lifecycle.

Artificial Intelligence and Privacy

GDPR Provision (The "What")

- **Art. 22:** Right to human intervention in automated decisions.
- **Art. 35:** Requirement for a Data Protection Impact Assessment (DPIA).
- **Art. 9:** Prohibition on processing sensitive data (e.g., race, religion)
- **Art. 5:** Principles of fairness, accuracy, and data minimization.

EU AI Act Provision (The "How")

- **Art. 14:** Obligation to build human oversight features into high-risk AI.
- **Art. 27:** Requirement for a broader Fundamental Rights Impact Assessment (FRIA), which complements the DPIA.
- **Art. 10(5):** A specific legal basis to process sensitive data *only* for bias detection and correction
- **Art. 10:** Specific rules on data governance to ensure datasets are relevant, representative, and error-free.

What is “AI system”?

Article 3(1)

- ‘AI system’ means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments;

Measures in support of innovation

AI regulatory sandboxes

- At least one per EU member nation
- A controlled environment that fosters innovation and facilitates the development, training, testing and validation of innovative AI systems
- Appropriate, guidance, supervision and support
- Guidance on regulatory expectations
- Handle personal data carefully
- Testing of high-risk AI systems in real world conditions outside AI regulatory sandboxes
 - Allowed but subject to constraints

Measures for providers and deployers, SMEs, start-ups

- Duty of member states
 - priority access to the AI regulatory sandboxes
 - organise specific awareness raising and training activities on the application of this Regulation
 - provide advice and respond to queries about the implementation of this Regulation
 - facilitate the participation of SMEs and other relevant stakeholders in the standardisation development process
- Take into account
 - specific needs of the SMEs and start-ups to be taken into account
 - fees for conformity assessment
 - reducing those fees proportionately to their size, market size and other relevant indicators
- AI office
 - provide standardised templates
 - develop and maintain a single information platform

Reporting of serious incidents

- Report to market surveillance authorities in provided duration

Penalties

- Non compliance of Article 5
 - Up to EUR 35,000,000 or 7% of global turnover, whichever is higher
- Others
 - Up to EUR 15,000,000 or 3% of global turnover
- Misinformation to authorities
 - Up to EUR 7,500,000 or 1% of global turnover
- Fines for providers of general-purpose AI models
 - Up to EUR 15,000,000 or 3% of global turnover

Link to my YouTube Channel

- <https://www.youtube.com/@KshitizVermaDL>
- Link to the EU AI Act playlist
 - <https://youtube.com/playlist?list=PLxlkzujLkmQ9ivaExAQVCe0IYzKzwedGn&si=Z0LWXzswxjl o317m>
- Or scan the QR code



Thank you!

Some exercises

Some questions!

- AI used in pharmaceutical research and development
- LLMs to discover new drugs
- AI for emergency healthcare patient triage systems
- AI for decisions on dispatching medical aid
- LLMs to take clinical notes
- Social scoring of individuals for health benefits?
- AI based systems used for administration in healthcare
- AI for risk assessment and pricing for health insurance
- AI for evaluating and classifying emergency medical calls
- AI chatbots providing advice on wellbeing
- AI generated medical deepfakes
- LLMs to generate synthetic patient data